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Mechanistic Interpretability of Emotion Inference in Large Language Models

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Abstract

Large language models (LLMs) show promising capabilities in predicting human emotions from text. However, the mechanisms through which these models process emotional stimuli remain largely unexplored. Our study addresses this gap by investigating how autoregressive LLMs infer emotions, showing that emotion representations are functionally localized to specific regions in the model. Our evaluation includes diverse model families and sizes, and is supported by robustness checks. We then show that the identified representations are psychologically plausible by drawing on cognitive appraisal theory—a well-established psychological framework positing that emotions emerge from evaluations (appraisals) of environmental stimuli. By causally intervening on construed appraisal concepts, we steer the generation and show that the outputs align with theoretical and intuitive expectations. This work highlights a novel way to causally intervene and control emotion inference, potentially benefiting safety and alignment in sensitive affective domains. *Code at:* `GitHub repo.`

1 Introduction

Large Language Models (LLMs) demonstrate remarkable capabilities in emotion recognition and reasoning tasks, occasionally surpassing human performance (Elyoseph et al., 2023; Tak and Gratch, 2024). Prior research primarily engages with LLMs as black boxes, utilizing zero-shot inference or in-context learning to gauge their performance on tasks such as emotion classification (Yongsatianchot et al., 2023; Broekens et al., 2023), emotional decision-making and situational appraisal (Tak and Gratch, 2023), emotional intelligence (Wang et al., 2023b), emotional dialogue understanding (Zhao et al., 2023), and generation of emotional text (Gagne and Dayan, 2023). However, there remains a limited understanding of *how* LLMs internally represent and process emotional information. Given LLMs' increasingly significant societal impact—spanning domains such as mental health (Sharma et al., 2023) and legal decision-making (Lai et al., 2024)—investigating these internal mechanisms is crucial.

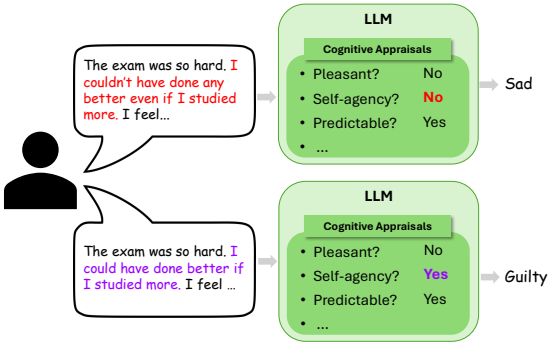

Figure 1: Emotion inference through latent appraisal-like mechanisms in LLMs. Given the description of a situation, the model leverages internal *appraisal* structures to recognize the emotion inferred from the context. For example, different perceptions of *self-agency* can distinguish between *guilt* and *sadness*.

Cognitive neuroscience uses functional localization approaches to identify specific brain regions responsible for particular functions and manipulate them by up/down-regulating neural activations in those regions. Akin to the shift from behaviorism to cognitive neuroscience in psychology—i.e. from treating the mind as a black box to studying brain-based cognitive processes—Mechanistic Interpretability (MI) allows for moving from black-box techniques (Casper et al., 2024), to a focus on the internal mechanics of LLMs (Bereska and Gavves, 2024). MI can offer a fundamental understanding of information processing and representation in LLMs, yielding insights into their inner-workings and offering new ways to control their reasoning (Li et al., 2021; Rai et al., 2024; Feng et al., 2025). Building on this line of research and by drawing inspiration from emotion theory in psychology, our work elucidates the inner workings of emotion processing in LLMs.

We start by training linear classifiers on top of hidden representations to probe for regions where the strongest emotion-related activations occur. We provide evidence for functional localization of emotion processing and show that emotion-relevant operations are concentrated in specific layers, a consistent behavior across various model families and scales. We complement these findings by applying causal interventions, namely patching activations in the computation graphs, to identify essential components in neural representations (Conmy et al., 2023). As a result, we show that Multi-Head Self-Attention (MHSA) units in the mid-layers are predominantly responsible for shaping the LLM decision. To further corroborate this, we visualize attention patterns, revealing that MHSA units consistently attend to emotionally loaded tokens. Our findings are robust and not influenced by variations in prompt wording or formatting.

Additionally, we use the *appraisal theory* from psychology to shed light on the structure of LLMs' internal representations. According to appraisal theory (Frijda et al., 1989; Scherer et al., 1984; Smith and Ellsworth, 1985), people reason about emotional situations by forming *appraisal* judgments (see Figure 1 as an example). We analyze the structure of emotion representations in LLMs by conducting inference-time probing on appraisal concepts to show that representations are psychologically plausible. Moreover, by modulating latent appraisal concepts to promote/demote a particular appraisal dimension (e.g., promoting *self-agency*), we show that the resulting changes in the output emotion align with theoretical expectations (from *guilt* to *sadness*) (Marcinkevičs et al., 2024; Wu et al., 2024; Li et al., 2023).

Overall, our work extends existing MI methodologies by applying them to more ecologically valid, unstructured examples, moving beyond the common practice of analyzing simplified sentence structures (Wang et al., 2023a; Hanna et al., 2023). Furthermore, this work serves as an early step to bridge MI techniques with applications in psychological and cognitive domains, offering insights into the inner workings of LLMs in complex, socially relevant contexts.

2 Related Work

Appraisal Theory. Appraisal theory is a model that explains how peoples' emotions are a result of their evaluations of a situation (Lazarus, 1991). It provides a comprehensive framework for understanding the precursors of emotions (Smith and Kirby, 2011). Neuroscience studies build on this framework by manipulating cognitive appraisals and examining associated brain activity, linking specific brain regions to appraisal processes (Leitão et al., 2020; Kragel et al., 2024; Brosch and Sander, 2013). These methods can be extended to evaluate how LLMs understand emotions, and identify the mechanisms responsible for those evaluations.

Cognitive-Neuroscientific Alignment. There is a growing body of evidence demonstrating that neural network activations can reflect cognitive constructs traditionally studied in psychology and neuroscience. For example, transformer self-attention mechanisms are shown to correlate with human eye-tracking data during reading, suggesting that LLMs may learn attention patterns that reflect human cognitive attention (Bensemann et al., 2022; Eberle et al., 2022). LLMs also exhibit strong representational alignment with neural activity in the human language systems, indicating convergence between learned representations and brain-like processing (Aw et al., 2024), and word embeddings show alignment with fMRI activity during word association tasks (Kwon et al., 2024). Recent work even uses LLM-derived representations to predict neural activation patterns tied to high-level semantic categories like faces or places, offering scalable proxies for annotation in cognitive experiments (Liu et al., 2025). Other studies localize LLM units that align with cortical language areas (AlKhamissi et al., 2025) and reveal functional specialization patterns predictive of regional brain activity (Kumar et al., 2024).

These findings support the plausibility of mapping psychological or cognitive theories—such as appraisal theory—onto the internal structure of LLMs. Our work builds on this foundation by examining how appraisal mechanisms are internally encoded, bridging prior behavioral studies of appraisal-emotion mappings in LLMs (Tak and Gratch, 2024, 2023; Broekens et al., 2023; Yongsatianchot et al., 2023) with MI.

Mechanistic Interpretability. Probing is an MI technique that uses a simple model, called a "probe", to assess the internal representations across various layers in a model. As explained by Belinkov (2018), the groundwork for what we now refer to as probing relates back to earlier work evaluating trained classifiers on static word embeddings to predict linguistic features (Köhn, 2015;

Gupta et al., 2015), and classified hidden states of neural models (Ettinger et al., 2016; Kádár et al., 2017; Shi et al., 2016; Adi et al., 2017; Hupkes and Zuidema, 2018; Belinkov, 2022; Giulianelli et al., 2018). Probing is used across a variety of tasks (Hewitt and Liang, 2019; Tenney et al., 2019a,b; Peters et al., 2018; Clark et al., 2019; Belinkov, 2018; Conneau et al., 2018).

Activation patching (Heimersheim and Nanda, 2024), is a causal intervention used to identify if certain activations are important to the downstream task (Vig et al., 2020). By using patching, Meng et al. (2022) are able to localize where models store factual information. Patchscope, a method that extends on activation patching, is used to translate LLM representations into natural language (Ghandeharioun et al., 2024).

Yet another MI technique is generation steering. This method entails manipulating a model's activations to control the outputs (Rai et al., 2024; Todd et al., 2024). Geva et al. (2022) highlight the role of Feed-Forward Network (FFN) units in promoting concepts. To steer generation, they apply sub-updates promoting safety and are able to reduce the model's toxicity. Templeton et al. (2024) find that "clamping" on features can be used to control the models' output, steering the model's stated goals and biases for both desirable and undesirable outputs. Nanda et al. (2023) demonstrate that sequence models can have linear internal representations and that these representations can be used to manipulate the model's behavior. This method closely resembles those of Turner et al. (2023) and Lieberum et al. (2023).

3 Experimental Setup

Dataset and Prompt Design. We employ the crowd-enVENT dataset developed by Troiano et al. (2023), which comprises 6,800 emotional vignettes annotated with self-reported emotions among a list of 13 options and 23 self-rated appraisal variables, reflecting nuanced stimuli evaluations, including: *pleasantness/unpleasantness*, *self-agency/other-agency*, *predictability/suddenness*. Appendix A.1 presents more details on the dataset, including a detailed list of appraisal variables, along with the scales used for measurement.

To evaluate the model's ability to infer emotions from textual contexts, we design prompts that guide the model to predict the appropriate emotion as the next immediate output token, framing the task as a causal language modeling problem. Subsequently, we consider a classification problem and evaluate the model by inspecting the logits confined to the set of targeted emotion labels. The primary prompt template used in this study is shown in Figure 4.

Emotion attribution is inherently subjective, making it challenging to define a single ground truth label for each input, particularly given our fine-grained list of emotions. Thus, we focus on the correctly classified examples when inspecting each language model. In other words, we only analyze the samples for which the LLM and the human annotator agreed on the same label, totaling at least 2,700 samples among different language models (see Appendix A.3 for more details). This ensures that we assess tasks where the model performs reliably to understand the underlying mechanisms.

Model Architecture. To account for the impact of model scale and architectural variations, we evaluate a diverse set of model families and sizes, including Llama 3.2 1B Instruct and Llama 3.1 8B Instruct (Grattafiori et al., 2024), Gemma 2 2B Instruct and Gemma 2 9B Instruct (Team et al., 2024), OLMo 2 7B Instruct and OLMo 2 13B Instruct (OLMo et al., 2024), Phi 3.5 mini Instruct and Phi 3 medium-Instruct (Abdin et al., 2024), and Ministral 8B Instruct and Mistral 12B Nemo Instruct (MistralAI, 2024) (see Appendix A.2 for more details). Some detailed analyses, robustness tests, and appraisal concept interventions are exclusively conducted on Llama 3.2 1B to manage computational resources effectively.

4 Notations and Preliminaries

Prior research suggests that both MHSA and FFN units drive the generation in specific downstream tasks such as indirect object identification or concept promotion (Merullo et al., 2024; Geva et al., 2022). By examining activations immediately after these units, we aim to evaluate their respective contributions to emotion processing within each transformer layer. More formally, let $\mathbf{h}_t^{(l)} \in \mathbb{R}^d$ denote the hidden state vector at layer l and token index $t \in \{1, \cdots, T\}$, where d is the dimensionality of the model's hidden representations and T is the input sequence length. Then,

$$\mathbf{a}_t^{(l)} = \mathbf{MHSA}(\mathbf{h}_{1:t}^{(l-1)}),$$

$$\mathbf{m}_t^{(l)} = \mathbf{FFN}(\mathbf{h}_t^{(l)} + \mathbf{a}_t^{(l)}),$$